D DigestAnything

RALPHTHON SEOUL #2

SOLO BUILD

DigestAnything

Any source → **See it. Test it. Prove it.**

Paste a URL, YouTube link, or PDF → get 5 AI-powered learning experiences

Builder
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GitHub
jays0606/DigestAnything

You read a 30-page paper yesterday.
How much can you explain right now?

10%
retained from reading

90%
retained from explaining

"If you can't explain it simply, you don't understand it." — Richard Feynman

The problem isn't access. It's the last mile.

ACCESS IS SOLVED

MIT OCW. YouTube. arXiv. The best lectures in human history are **free**.

UNDERSTANDING IS NOT

Reading ≠ learning. Watching ≠ understanding. You need someone to ask: *"explain it back to me."*

THE BOTTLENECK

A professor can't ask 300 students to explain one by one. But AI can. **Every student. Every concept. Infinitely.**

What cognitive science says works:

👁️ **Dual Coding**
Visual + verbal together = 2x retention

🧠 **Active Recall**
Output > input. Brain must retrieve to remember.

🔄 **Spaced Repetition**
Review right before you forget = long-term memory

💪 **Desirable Difficulty**
Struggle = learning. The brain must sweat.

SOLUTION

Paste Anything. Understand Everything.

Paste URL → LLM analyzes → **See it** → **Test it** → **Prove it**

🗺️ Overview

Interactive mindmap. See the whole structure at once.

SEE IT — Dual Coding

❓ Quiz

10 MCQs. Wrong = explanation. Brain must output.

TEST IT — Active Recall

🃏 Cards

15 flashcards. Review right before you forget.

TEST IT — Spaced Repetition

🎙️ Podcast

2-host AI conversation. Learn on your commute.

SEE IT — Elaborative Interrogation

🎓 Tutor ★

"Explain this." Finds your gaps. Never gives the answer.

PROVE IT — Feynman Technique

💬 Chat

Ask anything about the source. Always there.

On-demand reinforcement

HOW IT'S BUILT

One LLM Call → Everything in Parallel

Any source → one analysis → all outputs simultaneously

① Source Input
URL, YouTube, PDF — any content type

↓

② LLM Analysis
Structured extraction: concepts, hierarchy, insights, analogies

↓

③ Parallel Generation

Mindmap | Quiz | Cards | Podcast

asyncio.gather() — all at once, not sequential

Stack

Frontend	**Next.js + Tailwind**
Backend	**FastAPI (async Python)**
AI	**LLM (text + TTS)**
Design	**Google Stitch**
Mindmap	**Markmap (interactive)**

Design: "Cognitive Sanctuary"

No borders. No clutter. Tonal depth instead of lines. A quiet space designed for thinking — not another generic AI dashboard.

RALPH

895 Lines of Spec. Zero Lines of Manual Code.

What I Wrote (the spec)

SPEC.md — 895 lines

Every prompt, API structure, build order, verification criteria, fallback strategy

VERIFIED-API.md — 504 lines

Every LLM call tested and passing before implementation started

17 Test Scripts — 1,500+ lines

API tests, browser E2E (Playwright), audio quality verification — all written before code

Then I walked away.

Ralph rule: don't touch the laptop while the agent works.

What the Agent Did

1. **Read spec** → generated implementation plan
2. **Built 4 rounds** — BE + FE, commit after each
3. **Ran tests → failed → read error → fixed → retested**
4. **Browser E2E** — Playwright tested every tab interaction

Inspired by:

Anthropic's Harness Design — **Generator-Evaluator** pattern. Separate the builder from the critic. Self-evaluation bias means code can't judge itself.

IMPACT

The Last Mile of Education

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Understanding is not.

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AI can.
Every student. Every concept. Infinite patience.

Who Benefits

Students
Paste lecture PDF → quiz yourself → tutor catches gaps → actually pass the exam

Developers
Paste technical blog → mindmap architecture → quiz edge cases → explain to teammates

Lifelong Learners
Paste YouTube talk → podcast for commute → flashcards for spaced review

Educators
Generate quiz + cards from any material instantly. AI tutor handles 1-on-1.

PROGRESS

Current Status

AGENT RUNTIME

5h+

Autonomous build time

COMMITS

20+

All agent-authored

LINES OF CODE

5K+

Python + TypeScript

Overview	Quiz	Cards	Podcast	Tutor
Done	Done	Done	Done	Polishing

Human contribution: SPEC.md (895 lines) + test definitions. Everything else: autonomous agent.